MUKESH SEERAPU

Senior Data Engineer

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PROFESSIONAL SUMMARY

Extensive data engineering experience (10+ years) creating end-to-end data platforms within banks, financial services, and ecommerce. Expert level Azure Databricks (Delta Lake, Unity Catalog, Delta Live Tables, MLflow, Auto Loader) and skills within the Azure ecosystem (ADF, ADLS Gen2, Azure Synapse, Event Hubs). Ability to engineer Lakehouse architectures utilizing the Medallion Pattern (Bronze/Silver/Gold), design and build metadata-driven ingestion frameworks, implement CDC and SCD Type 1/2 pipelines, and automated data quality and observability systems. Experienced with AWS for hybrid and multi-cloud systems (S3, Glue, EMR, Redshift, Kinesis, Athena). Strong engineering skills with Python, PySpark and experienced with the optimization and tuning of Spark for large scale data engineering and machine learning. Experienced with Airflow, dbt on Databricks & Snowflake, CI/CD, and best practices of engineering with regulated financial data.

CORE COMPETENCIES

Databricks Lakehouse (Delta Lake, DLT, Unity Catalog, MLflow) | Azure Data Architecture (ADF, Synapse, ADLS Gen2, Event Hubs) | Python & PySpark Engineering | Spark Performance Tuning | Medallion Architecture (Bronze/Silver/Gold) | CDC & Incremental Loads | SCD Type 1/2 | Batch & Streaming Pipelines | Data Quality & Observability | Data Lineage | Airflow Orchestration | dbt on Databricks | CI/CD for Data Pipelines | Financial Data Governance | Dimensional Modeling | Agile / Scrum

TECHNICAL SKILLS

Databricks: Delta Lake, Delta Live Tables (DLT), Unity Catalog, Databricks Workflows, Auto Loader, Structured Streaming, MLflow, Databricks SQL, Photon Engine, Cluster Optimization, Databricks Repos

Microsoft Fabric: OneLake, Fabric Lakehouse, Fabric Data Engineering, Fabric Pipelines, Fabric Warehouse, Fabric Notebooks, DirectLake, Power BI Integration

Programming: Python (advanced), PySpark, SQL (advanced), Spark SQL, Scala, T-SQL, PL/SQL, Pandas, NumPy

Azure: Azure Databricks, Azure Data Factory, ADLS Gen2, Azure Synapse Analytics, Azure Blob Storage, Azure SQL Database, Event Hubs, Key Vault, Azure DevOps, Azure Monitor

AWS: S3, Glue, EMR, Redshift, Lambda, Kinesis, Athena, DynamoDB, RDS, EC2, CloudWatch, IAM, Secrets Manager, SageMaker

Data Engineering: Apache Spark, Spark Streaming, Kafka, Hadoop, Hive, Parquet, JSON, ORC, CDC, SCD Type 1/2, Medallion Architecture, Dimensional Modeling

ETL / ELT & Orchestration: Apache Airflow, dbt (Databricks & Snowflake), Azure Data Factory, AWS Glue Workflows, Talend, Informatica, SSIS, Snowpipe

Data Quality & Observability: DLT Expectations, Delta Lake Constraints, dbt Tests, Automated Reconciliation, Data Lineage Tracking, Azure Monitor Alerting, Schema Validation

Warehouses & Databases: Delta Lake, Snowflake, Azure Synapse, Amazon Redshift, PostgreSQL, Amazon RDS, SQL Server, MySQL, DynamoDB, Cassandra

DevOps & Tooling: Git, GitHub, Bitbucket, Jenkins, Azure DevOps, CI/CD, Jira, Confluence, Agile, Scrum

BI & Visualization: Power BI (DAX), Databricks SQL Dashboards, Tableau, Qlik Sense, Qlik Cloud

PROFESSIONAL EXPERIENCE

Senior Data Engineer | Dupaco Credit Union — Dubuque, IA (09/2024 – Present)

- Designed and built a banking data platform using Azure Databricks to ingest core banking, lending, cards, digital channel and vendor data and load it to ADLS Gen2 as a medallion lakehouse (bronze/silver/gold delta layers). The platform provides analytics, compliance and operational reporting.
- Built Delta Live Tables pipelines and enforced Delta constraints and DLT expectations at each layer for ACID-safe concurrent reads and idempotent writes on regulated financial data. Previously, custom ETL and Talend jobs were used for pipelines, which have now been replaced with self healing pipelines with lineage tracking.
- leveraged the Databricks Auto Loader for scalable, schema-evolving ingestion of ingest vendor files, core banking extracts and API payloads with configuration driven control totals and auditing with exceptions quarantined
- Developed reusable ingest applications using Python and PySpark for customer, account and loan domains to ingest and process huge volumes of data, with the history of SCD Type 1 and 2, CDC merge, and late-arriving record handling.
- Improved nightly data processing windows for financial datasets, by implementing various enhancements including broadcast joins, partition pruning, skew handling and adaptive query execution. I used the Photon engine for increased performance.
- Configured Unity Catalog for centralized governance across Databricks workspaces fine-grained access controls, column level security and lineage tracking to fulfill regulatory requirements of the credit unions
- Used ADF and Talend to migrate credit union data from on-premise to the cloud and Snowflake. I eliminated redundant cloud and data storage infrastructure. This reduced the latency of data ingestion and storage costs.

- Used dbt to build a tested dimensional model over gold delta layers, and integrated the model into Power BI via DirectLake using Microsoft Fabric. I ensured secured processing via Azure Key Vault, RBAC, encryption, and CI/CD via Databricks Repos and Azure DevOps

Technologies: *SQL, Azure Databricks, Delta Lake, Delta Live Tables, Unity Catalog, Auto Loader, MLflow, dbt, Microsoft Fabric, OneLake, DirectLake, Azure Data Factory, ADLS Gen2, Azure Key Vault, Azure DevOps, Snowflake, Power BI, Python, PySpark, Airflow, CI/CD,Postman*

Senior Data Engineer | Global Atlantic Financial Group — New York, NY (08/2023 – 09/2024)

- Implemented Databricks on AWS for processing of annuity and investment accounting data, replacing ad-hoc Spark processing on EMR clusters with a controlled and governed environment.
- Developed and implemented Delta pipelines in PySpark for transforming data (policy, transaction, valuation, and reference data) to curated Delta Lake Tables stored in S3 and consumed by Redshift and BI Tools. Delta pipelines were implemented to ensure audit trails and point-in-time reproducibility.
- Leveraged Databricks Streaming to provide near real-time data for actuarial reporting tables. Prior to this implementation, batch jobs for processing data for actuarial reporting left some tables unprocessed, leaving reporting incomplete.
- Integrated MLflow and Databricks Workflows to track experiment runs and govern model artifacts, as well as orchestrate workflows for ML projects (ingest → transform → validate → serve).
- Implemented Airflow to schedule dependent batch tasks for 30+ processes for data validation, retries, and notifications of failures. Additionally, implemented CI/CD pipelines for automating model and pipelines promotion.

Technologies: *SQL, AWS Databricks, Delta Lake, Structured Streaming, MLflow, Databricks Workflows, Databricks Repos, PySpark, S3, EMR, Redshift, Kinesis, CloudWatch, Apache Airflow, Jenkins, CI/CD, Tableau*

Data Engineer | Federal Home Loan Bank of Des Moines — Des Moines, IA (01/2023 – 08/2023)

- Built ELT pipelines in Azure Data Factory to move data from 15+ source systems into Azure Data Lake Storage Gen2 (ADLS Gen2), then processed the data using Spark and Snowflake as the serving warehouse.
- Designed an Azure Databricks lakehouse with a Bronze layer (raw data), a Silver layer (cleansed data with regulatory and credit datasets) and a Gold layer (business data) consumed by Power BI and SSRS.
- Ported legacy SSIS packages to Databricks notebooks, Snowflake-native SQL and Airflow to implement Snowpipe and SCD Type 2, remove the need for aging ETL infrastructure.
- Used Delta Lake in conjunction with Apache Airflow to provide time travel and lineage for data processed through the ADF and Databricks pipelines to meet regulatory audit requirements.
- Created validated Power BI and SSRS dashboards to visualize the KPIs for Finance and Risk stakeholders aligned with the reports defined by business teams.

Technologies: *SQL, Azure Databricks, Delta Lake, Azure Data Factory, ADLS Gen2, Azure Synapse Analytics, PySpark, Apache Airflow, Snowflake, Snowpipe, SQL Server, Informatica, SSIS, SSRS, Power BI*

Senior Data Engineer | OYO — Global Hotel Booking Platform — India (05/2018 – 11/2021)

- Designed reusable batch and streaming pipeline frameworks using Apache Spark, Kafka, Airflow/Oozie, and Hive, standardizing ingestion, data quality, and lineage controls across 10+ product teams supporting 5x growth in booking volume.
- Developed PySpark and Spark SQL applications for large-scale transformation of booking, pricing, customer, and clickstream data, and built real-time Spark Streaming pipelines for booking and behavior events.
- Integrated AWS SageMaker batch-inference outputs into downstream pipelines and CRM workflows, and applied storage platform selection (HDFS, Cassandra, Hive) based on data sensitivity and access patterns.

Technologies: *Python, PySpark, Spark SQL, Spark Streaming, Kafka, AWS S3, EMR, SageMaker, Apache Airflow, Hadoop, HDFS, Hive, Scala*

Engineering Lead & Co-Founder | Jhink Solution Pvt Ltd — India (01/2016 – 05/2018)

- Founded and led engineering for a B2B SaaS platform serving 200+ retail clients — designed the full data architecture from operational MySQL through Informatica-based ETL to analytics and reporting.
- Built REST APIs for real-time data consumption and Python ETL modules with built-in quality validation, maintaining 98% data accuracy while leading a 12-person engineering team through full SDLC.

Technologies: *Python, Django, MySQL, Informatica, REST APIs, JavaScript, Git*

Database Developer | Axis Bank — India (05/2015 – 12/2015)

- Developed and tuned PL/SQL procedures, functions, triggers, views, indexes, and constraints for high-volume banking transaction and reporting workloads, and partnered with ETL/SSIS developers to reconcile integration outputs.

Technologies: *PL/SQL, SQL Server, SSIS, Oracle*

EDUCATION

M.S. in Management Information Systems | Iowa State University | Iowa, USA (01/2022 – 05/2023)

MBA in Business Administration | Institute of Public Enterprise | India (08/2013 – 05/2015)

B.E. in Computer Science | JNTUK | India (08/2008 – 05/2012)

CERTIFICATIONS

- Microsoft Certified: Fabric Analytics Engineer Associate
- Databricks Lakehouse Fundamentals
- Business Analyst Certification — Iowa State University